Dr Sakshi Choudhary

Anne Bernheim's Laboratory, Department of Chemical Engineering
Ben-Gurion University of Negev, Israel

Curriculum Vitae

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Research Interest

- Biophysics, active matter, optical instrumentation, biomedical optics, spectroscopy etc.

Research Experience

- **Post-doctoral Research Fellow (January 2022 - continuing):**
  Department of Chemical Engineering
  Ben-Gurion University of Negev, Israel
- **Senior Research Fellow:** Biophysics Laboratory, Indian Institute of technology (ISM), Dhanbad, India (July 2017 – November 2020)
- **Junior Research Fellow:** Biophysics Laboratory, Indian Institute of technology (ISM), Dhanbad, India (July 2015 – July 2017)

Academic Qualifications

- **PhD in Physics** (November 2020)
  Indian Institute of Technology (Indian School of Mines), Dhanbad
  Thesis title: Nonlinear Photophysical Study of Biologically Active Molecules
  Supervisor: Dr Umakanta Tripathy
- **MSc in Physics** (2015)
  Lovely Professional University, Phagwara, Punjab, India
  Awarded with distinction, University Honor Roll and University Academic Honor
- **BSc, Physics & Mathematics** (2012)
  Vardhman College, Bijnor, UP, India
  MJP Rohilkhand University, Bareilly, India

Fellowships/Grants and awards

- Received international travel grant from IIT(ISM), Dhanbad to present research paper in the international conference icOPEN-2019, Phuket, Thailand July 2019
- Senior Research Fellowship from IIT(ISM), Dhanbad, July 2017-November 2020
- Junior Research Fellowship from IIT(ISM), Dhanbad, July2015-July 2017.
- Master's degree awarded with distinction, University Honor Roll and University Academic Honor for standing first in the order of merit
- Qualified prestigious IIT-JAM exam 2013
- Haryana government merit scholarship, 2009

Professional activities/membership

- SPIE, student member.
- Optical Society of America, student member.

Software and computer skills

- Operating system: Windows (XP, 7, 8, 10)
- Programming Language & software: C, MATLAB, Origin, imageJ
- Documentation: MS office
- Other software: CLP, Gaussian, SimphoSOFT

Training schools/internships

- National Fluorescence and Raman Spectroscopy Workshop: FCS 2016, at TIFR, Mumbai.
- Recent Trends in Photonics Technology: RTPT-2017 at IIT(ISM), Dhanbad.

Publications details

- **Research Highlighted in the media**

  Our scientific research published in the journal ACS Chemical Neuroscience is covered by Government of India Press Release and several other Indian media.

  - Press Information Bureau, Government of India
  - India Science Wire
  - India Education Diary
  - BioVoice News
  - Delhi Post News
  - EE Times India
  - The Kashmir Convener
  - Dainik Bhaskar
  - Indus Scrolls
  - Jatinverma.org

Papers published in peer reviewed science journals

1. V.C. Mandapati, H. Vardhan, S. Prabhakar, **Sakshi**, R. Kumar*, S.G. Reddy, R.P. Singh, K. Singh, Multi-user asymmetric optical cryptosystem based on polar de-composition and fractional vortex speckle patterns," Photonics, 10, 561, **2023 (IF = 2.536)**
2. **Sakshi**, Gefen Livne, Shachar Gat, Anne Bernheim-Groswasser "The Mechanics of (Poro) elastic contractile actomyosin networks as a model system of the cell cytoskeleton", *JOVE - Journal of Visualized Experiment* (accepted) **2023**. **(IF = 1.4)**
3. Ravi Kumar, **Sakshi,** and Kehar Singh, "An assymetric optical cryptosystem based on radon transform for phase image encryption," *Asian Journal of Physics*, 31, A1-A12, **2022**.
4. Nitesh Pathak, Lata Sharma, **Sakshi**, Bijayalaxmi Panda, and Umakanta Tripathy, "Synthesis, characterization, and investigation of nonlinear property of Fly Ash-Red mud- Ag: A low-cost sustainable nanocomposite," *Optical materials*, 126, 112230, **2022**. **(IF = 3.754)**
5. **Sakshi**, B.C. Swain, A.K. Das, N.K. Pathak, and U. Tripathy, "Z-scan analysis and theoretical studies of dopamine under physiological conditions," *Spectrochimica Acta part A: Molecular and Biomolecular Spectroscopy*, 271, 120890, **2022**. **(IF=4.831)**
6. **Sakshi**, B. C. Swain, A. K. Das, N. K. Pathak and U. Tripathy, "Norepinephrine exhibits thermo-optical nonlinearity under physiological conditions", *Physical Chemistry Chemical Physics (Communications)*, 23, 23473-23477, **2021**. **(IF = 3.945)**
7. **Sakshi**, N.K. Pathak, B.C. Swain, and U. Tripathy, "Analyzing nonlinear trends in curcumin: A comparative study", *Optics & Laser Technology*, 121, 105822, **2020**. **(IF = 4.939)**
8. S. Ghosh*, **Sakshi***, B.C. Swain, R. Chakraborty, U. Tripathy, and K. Chattopadhyaya, "A

novel tool to investigate the early and late-stage aggregation of alpha-synuclein", *ACS Chemical Neuroscience*, 11, 1610 – 1619, **2020**. ***Equal contribution (IF=5.780)**

9. B.C. Swain, S.K. Mukherjee, J. Rout, **Sakshi**, P.P. Mishra, M. Mukherjee, and U. Tripathy, "A spectroscopic and computational intervention of interaction of lysozyme with 6- mercaptopurine", *Analytical and Bioanalytical Chemistry*, 412, 2565-2577, **2020**. **(IF=4.478)**
10. B.C. Swain, S. Subadini, J. Rout, **Sakshi**, P.P. Mishra, H. Sahoo, and U. Tripathy, "Biophysical study on complex formation between β-Lactoglobulin and vitamin B12", *Food Chemistry*, 312, 126064, **2020**. **(IF=9.231)**
11. **Sakshi**, B.C. Swain, A.K. Das, and U. Tripathy, "Probing third-order nonlinearity in serotonin: A Z-scan study", *Spectrochimica Acta part A: Molecular and Biomolecular Spectroscopy*, 223, 117319, **2019**. **(IF=4.831)**
12. J. Rout, B.C. Swain, **Sakshi**, S. Biswas, A.K. Das, and U. Tripathy, "A simulation study on the influence of energy migration and relative interaction strengths of homo- and hetero-FRET on the net FRET efficiency", *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 226, 117599, **2019**. **(IF=4.831)**
13. **Sakshi**, Bikash Chandra Swain, and Umakanta Tripathy, "Applications of Lasers in Biophotonics", *Materials Focus*, 5, 496-510, **2016**. (*Review Paper*)

Peer reviewed international conferences proceedings

14. V. C. Mandapati, S. Prabhakar, R. Kumar, S.G. Reddy, **S. Choudhary**, R.P. Singh, "An asymmetric optical cryptosystem using physically unclonable functions in the Fresnel domain," at international conference on "Holography meets Advanced Manufacturing" organized by ERA Chair CIPHR Group, Institute of Physics, University of Tartu, Estonia, Europe, 20th - 22nd February 2023 DOI: https://doi.org/10.3390/HMAM2-14124 *(Best poster award - 3rd position)*
15. J. Rout, B.C. Swain, S. Subadini, **Sakshi**, P.P. Mishra, H. Sahoo, U. Tripathy, "Investigating the Binding Interaction between β-Lactoglobulin and Vitamin B12: A Spectroscopic and Computational Approach," *Biophysical Journal*, 120(3), 205a, **2021**. **(IF=4.033)**
16. **Sakshi**, and U. Tripathy, "Nonlinear property of neurotransmitter dopamine probed by Z- scan technique", Proc. SPIE 11205, Seventh International Conference on Optical and Photonic Engineering (icOPEN 2019), 112052L (16 October 2019); https://doi.org/10.1117/12.2542163.

Presentations/talks in national/international conferences

17. N.K. Pathak, **Sakshi**, B. Panda, and U. Tripathy, "Nonlinear study of silver-coated fly ash/red mud composite (FA/RM-Ag) by Z-scan Technique," *National Symposium on Light - Matter Interactions* (NSLIMI 2019) Dec. 26, 2019, IIT Madras, 2019
18. **Sakshi**, and U. Tripathy, "Nonlinear study of serotonin by a single closed aperture (CA) Z-scan technique", International Conference on Fiber Optics and Photonics, IIT, Delhi, 12th – 15th December, 2018, TP022. (*Poster presentation*)
19. **Sakshi**, and U. Tripathy, "Nonlinearity study of Curcumin by Z-scan technique", 26th *DAE-BRNS National Laser Symposium* (*NLS-26*), BARC, Mumbai, 20 - 23 December, 2017, CP- 04-43, P-57. (*Poster presentation*)
20. **Sakshi**, and U. Tripathy, "Estimation of Effective Pixel Size in Image Correlation Spectroscopy", *National conference on liquid crystals (NCLC)*, IIT(ISM) Dhanbad, 7 - 9 December, 2016, P-60. (*Poster presentation*)
21. **Sakshi**, and U. Tripathy, "Aggregation state study by Image Correlation Spectroscopy", *IONS*,

International OSA Network of Students, IIT(ISM) Dhanbad, 7 - 10 September, 2016, IONS-DHN/2K16/122. (*Oral presentation*)

22. Participated in national conference on "*Exploring Basic and applied Science for Next Generation Frontiers (EBAS 2014)*" organized by Lovely Faculty of Technology and Science, LPU, Phagwara, Punjab, 2014.

Books/Book Chapters:

1. R. Kumar, Y. Xiong, and **Sakshi**, (2023) Optical Cryptosystems Based on Spiral Phase Modulation, In: Chen, H., Liu, Z. (eds) Recent Advanced in Image Security Technologies, Studies in Computational Intelligence, Volume 1079, 2023. Springer Cham. ISBN: 978-3-031-22808-7. DOI: https://doi.org/10.1007/978-3-031-22809-4_3